

III SEMESTER

SNo	Course Code	Course Title	Category	L	T	P	C
1	CSCI7141 CSCI7161	Generic Elective -II Server –Side Programming with Java OR Server –Side Programming with Python	GE	3	0	0	3
2	CSCI7181	Design and Analysis of Algorithms	PC	3	0	0	3
3	CSCI7191	Artificial Intelligence and Machine Learning	PC	3	0	0	3
4	CSCI7201 CSCI7211 CSCI7231	Generic Elective – III (a)Business Intelligence (b)Agile Software Development (c) Cloud Computing: Concepts and Technologies	GE	3	0	0	3
5	CSCI7251 CSCI7261 CSCI7271 CSCI7281 CSCI7291	Generic Elective – IV (a)Computer Networks (b) Data Mining: Concepts and Techniques (c) Introduction to Parallel Computing (d) Cryptography and Network Security (e) Digital Forensics	GE	3	0	0	3
6	CSCI7301 CSCI7311 CSCI7321 CSCI7331	Skill Enhancement Courses (a) Digital Marketing (b) E-Commerce Strategies and Technologies (c) Social Network Analysis (d)Compiler Design	SEC	3	0	0	3
7	CSCI7341	Machine Learning Lab	PP	0	0	2	1
8	CSCI7151 CSCI7171	Server-Side Programming with Java Lab OR Server-Side Programming with Python Lab	PP	0	0	2	1
9	CSCI7351 CSCI7221 CSCI7241	(a) Business Intelligence using Tableau/Power BI Lab (b) Agile Software Development Lab (c) Cloud computing Lab	PP	0	0	2	1
10	CSCI7361	Internship/Industrial Training and Seminar					2
		Total					23

SERVER –SIDE PROGRAMMING WITH JAVA

L	T	P	S	J	C
3	0	0	0	0	3

Course Description:

This course offers a comprehensive exploration of server-side programming using Java technologies, focusing on developing robust and scalable web applications. It delves into various Java-based frameworks and technologies, including Servlets, JavaServer Pages (JSP), Java Database Connectivity (JDBC), Hibernate, and Spring. Through a series of practical, hands-on labs and projects, students will learn to build, deploy, and manage web applications effectively, integrating both relational database management systems and ORM technologies.

Course Objectives:

- To understand and apply Java Servlet technology.
- To develop Dynamic Web Applications
- To implement Database Interactions
- To utilize ORM Techniques
- To leverage Spring Framework

UNIT-I

Introduction to server-side programming: Role of Java in server-side development, Differences between application server and web servers.

Setting up the Java development environment: IDE setup, and server configurations
Servlet lifecycle, Writing and deploying a simple servlet, servlet-to-servlet communication, session management , cookie management , URL Rewriting. 9 Hours

UNIT-II

Introduction to JSP: Basics of JSP, JSP syntax and directives, JSP implicit objects, Use of scriptlets, expression language, and JSTL (JavaServer Pages Standard Tag Library). Error handling in JSP , Cookies, HTTP session tracking, creating filters, and listeners. 9 Hours

UNIT-III

Introduction to JDBC: JDBC architecture, register driver , connecting to databases.

Performing CRUD operation: Using JDBC to create, read, update, and delete records.

Connection pooling and data source: Optimizing database connections.

Integrating JDBC with servlets and JSP: Building dynamic web applications using databases. 9 Hours

UNIT-IV

Basics of ORM and Hibernate Setup: Introduction to Hibernate, Concept of ORM, configuring Hibernate, session management.

Hibernate Operations: CRUD operations, querying with HQL, introduction to criteria API.

Associations and Caching: Mapping relationships, utilizing first-level and second-level caches. 9 Hours

UNIT-V

Spring Core: Dependency injection, bean lifecycle, application context configuration.

Spring MVC and Spring Boot: Setting up Spring MVC, understanding controllers, Spring Boot basics for rapid development.

Spring Data JPA: Integrating Hibernate with Spring, repository abstraction layer, transaction management.

9 Hours

Text Books :

- 1.Head First Servlets and JSP: Passing the Sun Certified Web Component Developer Exam Bryan Basham, Kathy Sierra, and Bert Bates, O'Reilly Media, 2nd Edition (2008)
- 2.Full Stack Java Development with Spring MVC, Hibernate, jQuery, and Bootstrap Mayur Ramgir,Kindle Edition,

Course Outcomes:

After completing the course student will be able to:

- **CO1:** Understand the fundamentals of server-side programming, including servlet lifecycle, session management, and servlet-to-servlet communication, to build foundational web applications. *(L2: Understand)*
- **CO2:** Apply Java Server Pages (JSP) features, including directives, implicit objects, expression language, and JSTL, to create dynamic and interactive web pages with error handling and session tracking. *(L3: Apply)*
- **CO3:** Implement JDBC to connect with databases, perform CRUD operations, and integrate it with servlets and JSP to build data-driven web applications. *(L3: Apply)*
- **CO4:** Evaluate Hibernate ORM features, including CRUD operations, HQL, and caching mechanisms, to optimize database interactions in web applications. *(L5: Evaluate)*
- **CO5:** Design and implement enterprise-level web applications using Spring Core, Spring MVC, and Spring Boot frameworks, integrating Hibernate and Spring Data JPA for transaction management and repository abstraction. *(L6: Design)*

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	3	2	1	0	2
CO2	3	3	3	3	2	1	0	2
CO3	3	3	3	3	2	1	0	3
CO4	3	3	3	3	2	1	0	3
CO5	3	3	3	3	2	1	0	3

Note: 1 - Low Correlation 2 - Medium Correlation 3 - High Correlation

APPROVED IN:	
BOS : 8.6.2024	ACADEMIC COUNCIL:4.7.2024
SDG No. & Statement:	
4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.	
SDG Justification:	

The topics included in this course are designed to get acquainted with one of the skills that handle necessary mathematical orientation, programming techniques and concept based learning.

Prepared by Mr.B.Satyanarayana

Verified by Dr.M.Srivenkatesh

SERVER-SIDE PROGRAMMING WITH PYTHON

L	T	P	S	J	C
3	1	0	0	0	4

Course Description:

This course is designed to equip students with the skills necessary for full stack web development using Python, focusing on two of Python's most popular frameworks: Flask and Django. The course covers a broad spectrum of topics from basic Python programming for web applications to advanced features in Flask and Django, including RESTful API development, database integration, security implementations, and deployment strategies. Students will engage in hands-on projects that emphasize practical applications, enabling them to build and deploy dynamic web applications.

Course Objectives:

1. To master Python for Web Development
2. To develop Web Applications Using Flask
3. To implement Database Integration with SQLAlchemy
4. To explore Django's Capabilities
5. To deploy Django Applications

UNIT -I

Introduction to Python for Web Development

Python Basics: Quick overview of Python syntax, data types, control structures, and functions.

Setting up the Development Environment: Installing Python, virtual environments, and introduction to pip.

Introduction to Web Programming: Understanding client-server architecture, HTTP protocol basics. 9 Hours

UNIT-II

Flask Framework

Introduction to Flask: Setting up a Flask application, basic routing, and template rendering.

Flask Deep Dive: Handling forms, cookies, and sessions. Introduction to Flask blueprints for larger applications.

Database Integration: Connecting a Flask application to a database using SQLAlchemy ORM; performing CRUD operations. 9 Hours

UNIT -III

Advanced Flask Features

RESTful API Development: Building RESTful services with Flask-RESTful.

Middleware and Extensions: Utilizing Flask extensions like Flask-Mail for email handling, Flask-WTF for form validations.

Security Features: Implementing authentication (Flask-Login), and authorization, securing APIs.

Django Framework

UNIT –IV

Hours

Introduction to Django: Setting up a Django project, understanding the MVT pattern.

Models and Databases: Deep dive into Django ORM, model relationships, migrations.

Handling HTTP Requests: Creating views, class-based views, form handling, and admin interface.

9 Hours

UNIT –V

Building and Deploying Django Applications

Advanced Django Features: Middleware, custom template tags and filters, signals and slots.

Integrating Frontend Technologies: Using static and media files, integrating with JavaScript and AJAX for dynamic content.

Deployment: Deployment strategies for Django applications, using services like Heroku, understanding Docker for deployment.

9 Hours

Text Books :

1. Flask Web Development: Developing Web Applications with Python by Miguel Grinberg
Python Web Programming with Flask by Fabrizio Romano, O'Reilly Media, 2nd Edition, 2018
2. Django for Beginners: Build websites with Python and Django by William S. Vincent,
3rd Edition (2020)
3. Two Scoops of Django 3.x: Best Practices for the Django Web Framework by Audrey Roy
Greenfeld and Daniel Roy Greenfeld, Two Scoops Press, 1st Edition, 2020
4. Django 3 By Example by Antonio Melé, Packt Publishing, 3rd Edition, 2020

Course Outcomes:

After completing the course student will be able to:

- **CO1:** Understand Python basics, web programming concepts, and the client-server architecture to set up a development environment for web applications. *(L2: Understand)*
- **CO2:** Apply Flask framework features, including routing, form handling, and database integration using SQL Alchemy ORM, to build web applications. *(L3: Apply)*
- **CO3:** Analyse advanced Flask functionalities, such as RESTful API development, middleware usage, and security features, to implement scalable and secure web services. *(L4: Analyse)*
- **CO4:** Evaluate Django framework concepts, including MVT patterns, ORM features, and class-based views, to manage complex web application workflows. *(L5: Evaluate)*
- **CO5:** Create and deploy full-stack Django applications, integrating frontend technologies, custom features, and deployment strategies like Docker and Heroku. *(L3: Design)*

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	2	3	2	0	0	3
CO2	3	3	3	3	2	0	0	3
CO3	3	3	3	3	2	0	0	2
CO4	3	3	3	3	2	0	0	3
CO5	3	3	3	3	2	0	0	3

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DESIGN AND ANALYSIS OF ALGORITHMS

L	T	P	S	J	C
3	0	0	0	0	3

Course Description:

This course is designed to provide students with a thorough understanding of algorithm design and analysis. It covers a wide range of algorithmic strategies including divide and conquer, greedy methods, dynamic programming, as well as basic traversal and advanced search techniques. The course emphasizes both the theoretical underpinnings of algorithms and their practical applications, using classical problems like the knapsack problem, job sequencing, and graph coloring to illustrate concepts. Students will also learn about NP-hard and NP-complete problems, equipping them with the tools to tackle complex computational problems.

Course Objectives:

- To understand Algorithmic Foundations
- To learn about Divide and Conquer Techniques
- To apply Greedy Methods
- To explore Dynamic Programming
- To implement Advanced Search Techniques

UNIT – I

Introduction To Algorithms: Algorithm Specification, Performance Analysis, Introduction To Randomized Algorithms. Divide And Conquer: The General Method, Binary Search, Finding Maximum & Minimum, Quick Sort, Selection, Strassen's Matrix Multiplication. 9 Hours

UNIT – II

Greedy Method: General Method, Knapsack Problem, Tree Vertex Splitting, Job Sequencing With Deadlines, Minimum Cost Spanning Trees, Single Source Shortest Paths. 9 Hours

UNIT – III

Dynamic Programming: The General Method, Multistage Graphs, All Pairs Shortest Paths, Optimal Binary Search Trees, String Editing, Reliability Design, the Traveling Sales Person Problem. 9 Hours

UNIT – IV

Basic Traversal And Search Techniques: Techniques For Graphs, Connected Components And Spanning Trees, Bi-Connected Components and DFs. **Back Tracking:** General Method, Eight Queens Problem, Sum Of Subsets, Graph Coloring, Hamiltonian cycles. 9 Hours

UNIT – V

Branch and Bound: The method, 0/1 knapsack problem, traveling salesperson problem algebraic problems: the general method, evaluation and interpolation. np hard and np complete problems: basic concepts. 9 Hours

Text Book:

1. Fundamentals of Computer Algorithms By Ellis Horowitz, Sartaj Sahni, Sanguthevar

Rajasekaran, University Press, 2nd Edition, 2008.

Reference Books:

1. Fundamentals of Algorithmics By G. Brassard And P. Bratley, Phi, 2011.
2. Introduction to Algorithms By T.H. Cormen, C.E. Leiserson, R.L. Rivest, 3rd Edition, PHI, 2010.
3. Introduction to Design Analysis of Algorithms By Anany Levitin, 2nd Edition, Pearson Publications, 2009.

Course Outcomes:

After completing the course student will be able to:

- **CO1:** Understand algorithm specification, performance analysis, and the divide-and-conquer paradigm to solve problems like binary search, quick sort, and matrix multiplication. (*L2: Understand*)
- **CO2:** Apply the greedy method to solve optimization problems such as knapsack, job sequencing, minimum spanning trees, and shortest path problems. (*L3: Apply*)
- **CO3:** Analyse dynamic programming techniques for solving complex problems, including multistage graphs, all-pairs shortest paths, optimal binary search trees, and the traveling salesperson problem. (*L4: Analyse*)
- **CO4:** Evaluate graph traversal techniques, backtracking methods, and their applications to problems like graph coloring, Hamiltonian cycles, and the eight queens problem. (*L5: Evaluate*)
- **CO5:** Develop solutions for complex computational problems using branch-and-bound methods and assess NP-hard and NP-complete problems in the context of algorithm design. (*L3: Develop*)

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	1	-	-	2	-	3
CO2	3	3	2	-	-	2	-	3
CO3	3	3	3	-	-	2	1	3
CO4	3	3	3	-	-	2	-	3
CO5	3	3	3	-	-	1	-	3

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Prepared by Dr.U.Santosh Kumar

Verified by Dr.M.Srivenkatesh

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

L	T	P	S	J	C
3	0	0	0	0	3

Course Description:

This course provides a comprehensive introduction to the field of Artificial Intelligence (AI) and its application in Machine Learning (ML). It covers a broad spectrum of topics, ranging from basic AI principles and search algorithms to advanced machine learning techniques and neural networks. Students will explore both the theoretical underpinnings and practical implementations of AI and ML, gaining insights into how these technologies can be leveraged to solve real-world problems.

Course Objectives

- To understand the Fundamentals of Artificial Intelligence:
- To explore Logic and Reasoning in AI
- To Learn Core Concepts of Machine Learning
- To develop Skills in Modelling and Evaluating AI Systems
- To gain Proficiency in Supervised and Unsupervised Learning Techniques

UNIT-I

Introduction to artificial intelligence: Introduction, foundations of AI, applications

Search Algorithms:

Uninformed –Dijkstra Algorithm, Depth Limited, Iterative deepening Search

Informed Search –Best First, A* and AO* Algorithms

9 Hours

UNIT-II

What is logic in AI, What is reasoning in AI, Types of reasoning in AI, Limitations of logic and reasoning in AI.

Introduction: What is machine learning, Types of Machine Learning- Supervised Machine Learning, Unsupervised Machine Learning, Reinforcement Learning, Problems not solved with Machine Learning, Applications of Machine Learning

Preparing to Model: Machine Learning Activities, Basis types of Data in Machine Learning, Exploring the structure of the data, Data Quality and Remediation, Data Preprocessing

9 Hours

UNIT-III

Basics of Future Engineering: Introduction ,Future Transformation ,Future Subset Selection

Supervised Learning1: Introduction, Examples of Supervised Learning, Classification Model, Classification Steps, Common Classification Algorithms KNN, Decision Tree, Random forest model, Support vector Machines.

9 Hours

UNIT IV

Supervised Learning2: Introduction, Examples of Regression, Common Regression Problems includes simple, Multi linear regression, Assumptions in Regression Analysis, Main Problems in Regression Analysis, Polynomial Regression, Logistic Regression

Modelling & Evaluation: Introduction, Selecting a model, Training a model, Model Representation and Interpretability, Evaluating Model Performance, Improving model performance. 9 Hours

UNIT V

Unsupervised Learning: Introduction, Unsupervised vs Supervised Learning, Application of Unsupervised Learning, Clustering, Clustering as a machine learning task, Different types of clustering techniques, Partitioning methods, K-Means, K-Medoids: a representative object-based technique, Hierarchical Clustering.

9 Hours

Basic Neural Networks: Neural Network, Understanding Biological Neuron, Exploring Artificial Neuron, Types of activation function, Early implementation of ANN, Architecture of Neural Networks, Learning process in Artificial Neural Networks, Back Propagation, Deep Learning. 9 Hours

Text Books:

1. Artificial Intelligence A Modern Approach, Stuart Russell, Peter Norvig, Person Edition, Fourth Edition.
2. Machine Learning by Subramanian, Chandra Mouli, Amit Kumar Das, Saikant Dutt, Pearson Publications, I edition, 2018.
3. Machine Learning by Tom Mitchell, McGraw Hill, 2007

Course Outcomes:

After completing the course student will be able to:

- **CO1:** Understand the concepts of logic and reasoning in AI, machine learning fundamentals, and data preprocessing techniques to prepare datasets for modeling. (*L2: Understand*)
- **CO2:** Apply feature engineering techniques and supervised learning algorithms, including KNN, decision trees, random forests, and SVM, to solve classification problems. (*L3: Apply*)
- **CO3:** Analyse regression techniques, including simple and multiple linear regression, polynomial regression, and logistic regression, to model and evaluate supervised learning problems. (*L4: Analyse*)
- **CO4:** Evaluate unsupervised learning methods, such as clustering, and explore partitioning and hierarchical clustering techniques to group data and uncover patterns. (*L5: Evaluate*)
- **CO5:** Develop and implement neural network models by understanding their architecture, activation functions, and learning processes, and apply backpropagation and deep learning concepts. (*L3: Develop*)

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	3	2	0	0	3
CO2	3	3	3	3	2	0	0	3
CO3	3	3	3	3	2	0	0	3
CO4	3	3	3	3	2	0	0	3
CO5	3	3	3	3	2	0	0	3

Note: 1 - Low Correlation 2 - Medium Correlation 3 - High Correlation

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Prepared by Dr.M.Srivenkatesh,
Dr.U.Santosh Kumar

Verified by Dr.M.Srivenkatesh

BUNINESS INTELLIGENCE

L	T	P	S	J	C
3	0	0	0	0	3

Course Description:

This course provides an in-depth exploration of 99999 (BI), encompassing its development, methodologies, technologies, and applications in various industry sectors. The curriculum covers foundational BI concepts, data management, and the use of agent-based systems, advanced time series analysis, and the strategic implementation of BI in marketing models. Students will gain practical insights into how BI tools and technologies are applied to enhance decision-making processes, optimize performance, and drive organizational strategy.

Course Objectives:

- To understand the Core Concepts of BI
- To master Data Management for BI
- To explore BI Technologies and Tools
- To implement and Manage BI Projects
- To apply Advanced Techniques and Technologies

UNIT-I

Introduction to Business Intelligence

Overview: Definition and Scope of Business Intelligence, Historical Evolution of BI, Key Components of BI Systems, Role of BI in Organizational Strategy

Data Management for BI

Data Warehousing Concepts: Architecture, OLAP, OLTP, Data Mining: Methods and Techniques, Data Quality, Data Integration, and Data Governance

BI Technologies

Introduction to BI Software and Tools, Overview of Reporting, Dashboards, and Visualization Techniques, Basic Analytical Methods Used in BI

Implementation Challenges

BI Project Management: Planning and Execution, Common Pitfalls in BI Implementation and How to Avoid Them.

9 Hours

UNIT-II

Agent Systems in Business Intelligence

Agent-Based Modeling for BI: Introduction to Agent-Based Models in BI, Characteristics and Capabilities of Intelligent Agents, Types of Agents: Reactive, Proactive, and Hybrid

Agent Communication: Languages and Protocols for Agent Communication, Ontologies in Agent Systems: Role and Implementation, Inter-Agent Communication and Cooperation

Agent Systems Architecture: Architecture of Multi-Agent Systems (MAS), Design and Deployment of Agent Systems, Case Studies: Use of Agent Systems in BI Applications

Applications of Agent Systems in BI: Real-time Data Collection and Analysis, Automated Decision-Making Processes, Enhancing User Experience and Interface with Intelligent Agents.

9 Hours

UNIT- III

Agent Technologies and Their Applications in Business Intelligence

Advanced Agent Technologies: Learning Agents and Adaptive Systems, Mobile Agents and Their Applications in BI, Security Aspects of Agent Technologies

Integrating AI with Agent Technologies: AI-Driven Agents for Predictive Analytics, Natural Language Processing for Enhanced Data Interaction, Machine Learning Models Deployed via Agents

Implementation and Management: Deployment Strategies for Agent-Based Systems Performance Evaluation and Scaling of Agent Systems, Maintaining and Updating Agent-Based Systems

Industry-Specific Applications

Financial Services: Risk Assessment and Fraud Detection, Healthcare: Patient Data Analysis and Management, Retail: Customer Behaviour Analysis and Personalization Strategies. 9 Hours

UNIT- IV

Time Series Data in Business Intelligence

Introduction to Time Series: Definition, Components of Time Series

Time Series Analysis: Moving Averages, Decomposition, Exponential Smoothing

Advanced Time Series Models: Autoregressive Models, Moving Average Models, ARMA and ARIMA Model, Forecasting with Time Series

Business Intelligence Applications: Marketing Models

Overview, Relational Marketing, Key Metrics and Strategies: Lifetime Value (LTV), Customer Acquisition and Retention, Cross-Selling and Up-Selling

UNIT- V

Business intelligence applications: Marketing models -Relational marketing, Motivations and objectives, An environment for relational marketing analysis, Lifetime value, The effect of latency in predictive models, Acquisition, Retention, Cross-selling and up-selling, Market basket analysis, Web mining, Business case studies, Retention in telecommunications, Acquisition in the automotive industry, Cross-selling in the retail industry.

9 Hours

Text Books :

1. Decision Support Systems for Business Intelligence by Vicki L. Sauter, Wiley Publisher, 2nd edition, 2010.
2. Business Intelligence by Rajiv Sabherwal, Irma Becerra-Fernandez, Wiley, 2010.
3. Business Intelligence: A Managerial Perspective on Analytics" by Ramesh Sharda, Dursun Delen, and Efraim Turban
4. Time Series Analysis and Its Applications: With R Examples" by Robert H. Shumway and David S. Stoffer

Course Outcomes:

After completing the course student will be able to:

- **CO1:** Understand the fundamental concepts of Business Intelligence (BI), including data management, BI technologies, and implementation challenges, to align BI strategies with organizational goals. **(L2: Understand)**
- **CO2:** Apply agent-based systems in BI to enhance real-time data collection, automated decision-making, and user experience through agent communication and multi-agent system architectures. **(L3: Apply)**
- **CO3:** Analyze advanced agent technologies and their integration with AI to improve predictive analytics, natural language processing, and machine learning applications in industry-specific BI scenarios. **(L4: Analyze)**
- **CO4:** Evaluate time series analysis techniques, such as ARIMA models and exponential smoothing, to forecast business trends and implement relational marketing strategies in BI. **(L5: Evaluate)**
- **CO5:** Design BI applications for industry-specific use cases, such as retention in telecommunications, acquisition in the automotive industry, and cross-selling in retail, to optimize marketing strategies. **(L6: Create)**

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	2	2	0	0	3
CO2	3	3	3	2	2	0	0	3
CO3	3	3	3	3	2	0	0	3
CO4	3	3	3	3	2	0	0	3
CO5	3	3	3	3	2	0	0	3

Note: 1 - Low Correlation 2 - Medium Correlation 3 - High Correlation

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Mr.B.Srinivasa Rao

Verified by Dr.M.Srivenkatesh

AGILE SOFTWARE DEVELOPMENT

L	T	P	S	J	C
3	0	0	0	0	3

Course Description: Agile software development practices enable customer centric software development with collaborative teamwork centred around people. This course elaborates agile development principles and techniques covering the entire software development process from problem conception through development, testing and deployment to equip the learner with practical software development methodology

Course objectives:

- To understand the agile concept and its importance in software development.
- To acquire complete knowledge on Extreme programming.
- To know complete modelling of agile processes on the XP environment.
- To acquire knowledge on Scrum.
- To familiar with Feature driven development

UNIT -I

Introduction: The Agile manifesto, Agile methods, XP: Extreme Programming, DSDM, SCRUM, feature- Driven Development, Test Driven Development, modelling misconceptions, agile modelling, tools of misconceptions, updating agile models. 9 Hours

UNIT -II

Extreme Programming: Introduction, core XP values, the twelve XP practices, about extreme programming, planning XP projects, test first coding, making pair programming work. 9 Hours

UNIT -III

Agile Modelling and XP: Introduction, the fit, common practices, modelling specific practices, XP, objections to agile modelling, agile modelling and planning XP projects, XP implementation phase. 9 Hours

UNIT –IV

Scrum: Scrum Framework, Agile Principles, Sprints, Requirements and User Stories, Product backlogs, Estimation and Velocity, Roles, Planning, Multi-level Planning, Release Planning, Sprint planning. 9 Hours

UNIT -V

Feature-Driven Development: Introduction, incremental software development, Regaining Control, motivation behind FDD, planning an iterative project, architecture centric, FDD and XP.

Test Driven Development: Unit Tests, Integration Tests, End-to-End Tests, Customer Tests.

Release Management: Version Control, Continuous Integration.

9 Hours

Text Books:

John Hunt, Agile Software Construction, 1st Edition, Springer, 2005

Craig Larman, Agile and Iterative Development: A Manager's Guide, Addison-Wesley, Pearson Education – 2004.

Pearson, Robert C. Martin, Juli, James Shore, Chromatic 2013, The Art of Agile Development, O'Reilly Media.

Elisabeth Hendrickson, Agile Testing, Quality Tree Software Inc 2008.

Reference Books:

Andrew Stellman, Jenifer Greene, Headfirst Agile, O'Reilly, 2017

Peggy Gregory, Casper Lassenius, Xiaofeng Wang Philippe Kruchten (Eds.), Agile Processes in Software Engineering and Extreme Programming, 22nd International Conference on Agile Software Development, XP 2021 Virtual Event, June 14–18, 2021, Proceedings, Springer

Peggy Gregory, Philippe Kruchten (Eds.), Agile Processes in Software Engineering and Extreme Programming – Workshops XP 2021 Workshops Virtual Event, June 14–18, 2021 Revised Selected Papers, 2021

Ian Somerville, Software Engineering, 10th edition, Pearson, 2016

Course Outcomes:

After completing the course student will be able to:

CO1: Understand the Agile manifesto, methodologies, and practices such as Extreme Programming (XP), SCRUM, and Test-Driven Development (TDD) to foster agile software development. (*L2: Understand*)

CO2: Apply core Extreme Programming (XP) principles and practices, including pair programming and test-first coding, to plan and execute agile projects effectively. (*L3: Apply*)

CO3: Analyse the integration of Agile Modelling with XP, including common and modelling-specific practices, to address objections and improve project planning and implementation. (*L4: Analyse*)

CO4: Evaluate the SCRUM framework, including roles, sprints, and backlog management, to plan and execute agile projects efficiently. (*L5: Evaluate*)

CO5: Design iterative and incremental software development projects using Feature-Driven Development (FDD) and implement Test-Driven Development (TDD) with proper release management techniques. (*L3: Design*)

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	3	3	3	3	0	0	3
CO2	2	3	3	3	3	0	0	3
CO3	2	3	3	3	3	0	0	3
CO4	2	3	3	3	3	0	0	3
CO5	2	3	3	3	3	0	0	3

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CLOUD COMPUTING: CONCEPTS AND TECHNOLOGIES

L	T	P	S	J	C
3	0	0	0	0	3

Course Description:

This course provides a comprehensive overview of cloud computing, a transformative computing paradigm that has reshaped how organizations deploy, manage, and scale their IT resources. It delves into various forms of computing such as grid, cluster, and distributed computing, and focuses extensively on cloud computing technologies. Students will explore the evolution, architecture, and service models of cloud computing, including Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS). The course also addresses key issues such as virtualization technologies, service management, cloud security, and the economic impacts of cloud adoption.

Course Objectives:

- Understand the Fundamentals of Cloud Computing
- Explore Service Models and Deployment Models
- Master Virtualization Technology
- Examine Cloud Service Management and Economics
- Study Cloud Security and Compliance

UNIT -I

Introduction , Overview of Computing Paradigm - Recent trends in Computing - Grid Computing, Cluster Computing, Distributed Computing, Utility Computing, Cloud Computing - Evolution of cloud computing - Cloud Computing (NIST Model) – Characteristics

Pros and Cons of Cloud Computing, Cloud computing vs. Cluster computing vs. Grid computing – Role of Open Standards - Cloud Computing Architecture - Cloud computing stack - Service Models (XaaS) - Deployment Models.

9 Hours

UNIT -II

Infrastructure as aService(IaaS) Infrastructure as a Service(IaaS) – Introduction- IaaS definition, virtualization, Different approaches to virtualization, Hypervisors, Machine Image, Virtual Machine(VM) - Resource Virtualization – Server, Storage, Network, Virtual Machine(resource) provisioning and manageability, storage as a service, Data storage in cloud computing (storage as a service) - Examples - Amazon EC2 - Renting, EC2 Compute Unit, Platform and Storage,pricing, customers – Eucalyptus.

9 Hours

UNIT -III

Platform asa Service(PaaS) Platform as a Service(PaaS) - Introduction - What is PaaS, Service Oriented Architecture (SOA) - Cloud Platform and Management – Computation, Storage – Examples – Google App Engine, Microsoft Azure, Salesforce.com, Force.complatform-

Software as a Service(PaaS)-IntroductiontoSaaS-Webservices-Web2.0- WebOS-Case Study onSaaS.

9 Hours

UNIT –IV

Service Management in Cloud Computing (9hrs) Service Management in Cloud Computing – Service Level Agreements (SLAs) - Billing & Accounting - Comparing

Scaling Hardware: Traditional vs. Cloud-Economics of scaling: Benefitting enormously- Managing Data-Looking at Data, Scalability& Cloud Services, Database& Data Stores in Cloud, Large Scale Data Processing.

9 Hours

UNIT –V

Cloud Security Cloud Security - Infrastructure Security - Network level security, Host level security, Application level security – Data security and Storage Data privacy and security Issues,

Jurisdictional issues raised by Data location-Identity &Access Management-Access Control-Trust, Reputation,Risk-Authentication in cloud computing, Client access in cloud, Cloud contracting Model, Commercial and business considerations.

9 Hours

Text Books:

1. Rajkumar Buyya, James Broberg, Andrzej M. Goscinski, Cloud Computing: Principles and Paradigms, Wiley, 2011.
2. RonaldL.Krutz, Russell DeanVines, Cloud Security: A Comprehensive Guide to Secure Cloud Computing, WileyIndia,2010.

Reference Books:

1. Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi, Mastering Cloud Computing, Mc Graw Hill Education,2013.
2. NikosAntonopoulos, LeeGillam, Cloud Computing: Principles, Systems and Applications, Springer, 2012. 3. Barrie Sosinsky,Cloud Computing Bible, Wiley-India,2010.

Course Outcomes:

After completing the course student will be able to:

CO1: Understand the evolution, architecture, and characteristics of cloud computing, including its comparison with other computing paradigms, to identify its benefits and challenges. (*L2: Understand*)

1. **CO2:** Apply concepts of Infrastructure as a Service (IaaS), including virtualization, resource provisioning, and data storage, to explore and utilize cloud-based infrastructures like Amazon EC2 and Eucalyptus. (*L3: Apply*)

CO3: Analyse the features of Platform as a Service (PaaS) and Software as a Service (SaaS), including their role in enabling cloud-based applications, to understand their use cases and management strategies. *(L4: Analyse)*

CO4: Evaluate service management practices in cloud computing, such as SLAs, billing, and scalability, to manage and optimize cloud services for large-scale data processing. *(L5: Evaluate)*

CO5: Design secure cloud computing environments by implementing security measures at the infrastructure, network, and application levels, addressing data privacy, identity management, and access control issues. *(L3: Design)*

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	3	2	0	0	3
CO2	3	3	3	3	2	0	0	3
CO3	3	3	3	3	2	0	0	3
CO4	3	3	3	3	3	0	0	3
CO5	3	3	3	3	3	0	0	3

Note: 1 - Low Correlation 2 - Medium Correlation 3 - High Correlation

APPROVED IN:	
BOS : 8.6.2024	ACADEMIC COUNCIL:4.7.2024
SDG No. & Statement:	
4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.	
SDG Justification:	
The topics included in this course are designed to get acquainted with one of the skills that handle necessary mathematical orientation, programming techniques and concept based learning.	

COMPUTER NETWORKS

L	T	P	S	J	C
3	0	0	0	0	3

Course Description:

This course provides a comprehensive introduction to the principles and practical aspects of data communications, network architectures, protocols, and standards, with a specific focus on the Internet and related technologies. It covers the foundational theory behind data communication systems and network protocols, as well as the implementation and operational aspects of network layers, transmission methods, and advanced networking concepts. Students will explore both the theoretical underpinnings and the practical applications of building, maintaining, and securing networks.

Course Objectives

- To understand Network Fundamentals and Protocols
- To explore Network Media and Switching Technologies
- To master Data Link Layer Operations and Ethernet Technologies
- To understand Advanced Networking Concepts
- To apply Knowledge to Transport and Application Layers

UNIT - I

Introduction: Data Communications, Networks, The Internet, Protocols and Standards.

Network Models: Layered Tasks, The OSI Model, Layers in the OSI Model, TCP/IP Protocol Suite, Addressing.

Digital Transmission: Transmission Modes.

9 Hours

UNIT - II

Transmission Media: Guided Media, Unguided Media-Wireless Transmission.

Switching: Circuit Switched Networks, Datagram Networks, Virtual, Circuit Networks, Structure of a Switch.

Data Link Layer: Error Detection and Correction- Introduction, Block Coding, Linear Block Codes, Cyclic Codes, Checksum.

9 Hours

UNIT - III

Data Link Control: Framing, Flow and Error Control, Protocols, Noiseless Channels, Noisy Channels, HDLC, Point-to-Point Protocol.

Wired LANs: Ethernet-IEEE Standards, Standard Ethernet, Changes in the Standard, Fast Ethernet, Gigabit Ethernet.

Connecting LANs, Backbone Networks and Virtual LANs: Connecting Devices, Backbone Networks, Virtual LANs.

9 Hours

UNIT - IV

Network Layer: Logical Addressing - IPv4 Addresses, IPv6 Addresses.

Internet Protocol: IPv4, IPv6.

Network Layer: Address Mapping, ICMP, IGMP, ICMPv6.

Delivery, Forwarding and Routing: Delivery, Forwarding, Unicast Routing Protocols, Multicast Routing Protocols.

UNIT - V

Transport Layer: Process to Process Delivery- UDP, TCP and SCTP.

Application Layer: Domain Name System-Namespace, Domain Name Space, Distribution of Name Space, DNS in the INTERNET, RESOLUTION, DNS Messages, Types of Records, Registrars, Dynamic Domain Name System(DDNS), Encapsulation.

9 Hours

Text Books: 1. Data Communications and Networking by Behrouz A Forouzan, TMH, 4th edition, 2007.

Reference Books:

1. Data and Computer Communications by William Stallings, Pearson Publications, 9th edition, 2011.
2. Computer Networks by Andrew S. Tanenbaum, Prentice Hall, 5 th edition, 2012.

Course Outcomes:

After completing the course student will be able to:

CO1: Understand the fundamentals of data communication, networking models, and digital transmission to explain the layered architecture and key standards in networking. *(L2: Understand)*

CO2: Apply knowledge of transmission media and switching techniques to evaluate error detection and correction methods for reliable data communication. *(L3: Apply)*

CO3: Analyze data link control protocols, Ethernet standards, and techniques for connecting LANs to design efficient and robust local area networks. *(L4: Analyse)*

CO4: Evaluate logical addressing schemes, Internet protocols (IPv4 and IPv6), and routing techniques to ensure efficient delivery and forwarding of data packets. *(L5: Evaluate)*

CO5: Develop and implement transport and application layer protocols, including DNS and DDNS, to enable seamless communication and resolution in distributed systems. *(L3: Develop)*

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	3	2	0	0	2
CO2	3	3	3	3	2	0	0	2
CO3	3	3	3	3	2	0	0	2
CO4	3	3	3	3	2	0	0	3
CO5	3	3	3	3	2	0	0	3

Note: 1 - Low Correlation 2 - Medium Correlation 3 - High Correlation

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SDG Justification:	
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Prepared by Mr.M.Suresh Kumar

Verified by Dr.M.Srivenkatesh

DATA MINING: CONCEPTS AND TECHNIQUES

L	T	P	S	J	C
3	0	0	0	0	3

Course Description:

This course offers a comprehensive introduction to Data Mining, a crucial component of modern data analysis used to discover patterns and relationships in large datasets. It covers a broad range of topics from the basics of data mining, understanding and preparing data, to advanced techniques in pattern recognition, classification, clustering, and outlier detection. Students will learn about the various methods and technologies used in data mining, including how data warehousing and Online Analytical Processing (OLAP) contribute to mining processes.

Course Objectives:

- To understand the Foundations of Data Mining
- To develop Skills in Data Analysis and Pre-processing
- To learn to Utilize Data Warehousing and OLAP for Data Mining
- To acquire Techniques for Mining Patterns and Building Models
- To analyse Cluster Analysis and Outlier Detection

UNIT – I

Introduction: What is Data Mining? Kind of data on which mining is done, Kinds of patterns can be mined, and Technologies used, Kinds of Applications targeted, Major Issues of Data mining.

Getting to Know Your Data: Data Objects and Attribute, Measuring Data Similarity and Dissimilarity. 9 Hours

UNIT - II

Data Pre-processing: An Overview, Data Integration, Data Reduction, Data Transformation and Data Discretization.

Data Warehouse and OLAP Technology for Data Mining: Data Warehouse basic concepts, Data warehouse modeling. 9 Hours

UNIT – III

Mining Frequent patterns: Associations and Correlations: Basic Concepts, Frequent Item set Mining Methods, Interesting Patterns and Pattern evaluation methods. 9 Hours

UNIT - IV

Classification: Basic Concepts, Decision Tree Induction, Bayesian Classification, Rule based Classification, Model Evaluation & Selection, Techniques to improve classification Accuracy. 9 Hours

UNIT - V

Cluster Analysis: Requirements, Overview of various clustering methods, Density based Methods, Grid-based Methods, and Evaluation of Clustering.

Outlier Detection: Types of Outliers & Challenges in outlier detection, Outlier detection methods, Statistical Approaches, Proximity based approaches, Cluster & Classification

Text Books:

1. Data Mining Concepts and Techniques by Jiawei Han, Micheline Kamber, and Jian Pei, Elsevier Publications, 3rd Edition, 2013.

Reference Books:

1. Introduction to Data Mining by Pang-Ning Tan, Michael Steinbach and Vipin Kumar, Pearson Education, Low Price Edition.
2. Insight to Data Mining Theory and Practice by K.P. Soman, Shyam Diwakar and V. Ajay, Prentice Hall of India, 2006.

Course Outcomes:

After completing the course student will be able to:

- **CO1:** Understand the fundamental concepts of data mining, including data types, patterns, and similarity measures, to explore and analyze datasets effectively. (*L2: Understand*)
- **CO2:** Apply data preprocessing techniques, including data integration, reduction, transformation, and discretization, and utilize data warehousing concepts for data organization and preparation. (*L3: Apply*)
- **CO3:** Analyse frequent pattern mining techniques, such as association and correlation methods, to discover interesting patterns in datasets. (*L4: Analyse*)
- **CO4:** Evaluate classification techniques, including decision trees, Bayesian methods, and rule-based systems, to build accurate and efficient predictive models. (**L5: Evaluate**)
- **CO5:** Develop clustering models and implement outlier detection methods using statistical, proximity-based, and classification-based approaches to uncover hidden structures and anomalies in data. (**L3: Develop**)

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	3	2	0	0	3
CO2	3	2	3	3	2	0	0	3
CO3	3	3	3	3	2	0	0	3
CO4	3	3	3	3	2	0	0	3
CO5	3	3	3	3	3	0	0	3

Note: 1 - Low Correlation 2 - Medium Correlation 3 - High Correlation

APPROVED IN:	
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SDG Justification:	
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Prepared by Dr.M.Seshashayee

Verified by Dr.M.Srivenkatesh

INTRODUCTION TO PARELLEL COMPUTING

L	T	P	S	J	C
3	0	0	0	0	3

Course Description:

This course offers a comprehensive exploration into the concepts, techniques, and technologies of parallel programming. It is designed to equip students with the knowledge and skills required to develop efficient parallel algorithms and applications using various parallel programming platforms. The curriculum covers a broad spectrum of topics, including the architectural considerations of parallel systems, principles of parallel algorithm design, performance analysis, the message-passing programming paradigm, and specific parallel algorithms for matrix operations and sorting.

Course Objectives:

- To understand Parallel Computing Platforms
- To learn Parallel Algorithm Design
- To analyse Parallel Program Performance
- To Utilize the Message-Passing Paradigm
- To implement Parallel Algorithms and Applications

UNIT -I

Introduction : Introduction–Motivation–Scope-Parallel Programming Platforms: Implicit Parallelism– Limitations of Memory System Performance - Dichotomy of Parallel Computing Platforms – Communication cost in Parallel Machines – Routing Mechanism for Interconnection Networks.

9 Hours

UNIT - II

Principles of Parallel Algorithm , Decomposition Techniques – Characteristics of Tasks and Interactions – Mapping Techniques for Load Balancing – Methods for containing Interaction overheads – Parallel Algorithm Models – Basic communication Operations: One-to-all Broadcast and All-to-one Reduction – Scatter and Gather – Improving the Speed of some communication Operations.

9 Hours

UNIT - III

Analytical Modelling of Parallel programs ,Sources of Overhead in Parallel Programs – Performance metrics for parallel systems – effect of granularity and data mapping on performance – scalability of parallel systems – Minimum analysis of parallel Programs – other Scalability Metrics.

9 Hours

UNIT - IV

Programming using Message Passing Paradigm, Principles of Message-Passing Programming – The Building Blocks: Send and Receive Operations – MPI: The Message Passing Interface – Topologies and Embedding -Overlapping Communication with Computation - Collective Communication and Computation Operations - Groups and Communicators. 9 Hours

UNIT - V

Parallel Algorithms and Applications, Dense Matrix Algorithms: Matrix-Vector Multiplication- Matrix- Matrix Multiplication-Solving a System of Linear Equations. Sorting: Issues in Sorting on Parallel Computers - Sorting Networks - Bubble Sort and its Variants – Quick sort - Bucket and Sample Sort - Other Sorting Algorithms. 9 Hours

Text Book(s): 1. Ananth Grama, Anshul Gupta, George Karypis, Vipin Kumar, Introduction to Parallel Computing, , Pearson, Second Edition,2003.

Reference Book(s):

1. Peter S. Pacheco, An introduction to Parallel Programming, Morgan Kaufmann, First Edition,2011. 2. Fayez Gebali, Algorithms and Parallel Computing, Wiley series,2011.

Course Outcomes:

After completing the course student will be able to:

- **CO1:** Understand the fundamental concepts of parallel programming platforms, including communication costs and interconnection networks, to identify the challenges and opportunities in parallel computing. **(L2: Understand)**
- **CO2:** Apply principles of parallel algorithm design, decomposition techniques, and load-balancing strategies to optimize parallel computation tasks. **(L3: Apply)**
- **CO3:** Analyse performance metrics, scalability factors, and sources of overhead in parallel programs to evaluate the efficiency of parallel computing systems. **(L4: Analyse)**
- **CO4:** Evaluate message-passing programming principles and MPI operations, including collective communication and topologies, to design efficient parallel programs. **(L5: Evaluate)**
- **CO5:** Design parallel algorithms for dense matrix computations, sorting techniques, and solving linear equations to solve computational problems on parallel systems. **(L3: Design)**

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	3	2	0	0	3
CO2	3	3	3	2	2	0	0	3
CO3	3	3	3	2	2	0	0	3
CO4	3	3	3	2	2	0	0	3
CO5	3	3	3	2	2	0	0	3

Note: 1 - Low Correlation 2 - Medium Correlation 3 - High Correlation

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CRYPTOGRAPHY AND NETWORK SECURITY

L	T	P	S	J	C
3	0	0	0	0	3

Course Description:

This comprehensive course in Network and Computer Security provides a detailed examination of the fundamental principles, techniques, and technologies used to secure computer systems and networks. It delves into the architectures, protocols, and standards that ensure the confidentiality, integrity, and availability of information systems. Covering a broad spectrum from symmetric and asymmetric cryptography to securing network communications and managing security threats, this course equips students with the knowledge and skills necessary to design, implement, and manage secure computer systems.

Course Objectives:

- To understand Fundamental Concepts of Computer Security
- To analyze Cryptographic Techniques and Protocols
- To explore Security at Different Network Layers
- To implement Security in Wireless Networks and Electronic Communications
- To identify and Mitigate Threats from Malicious Software

UNIT- I

Introduction: Computer Security Concepts, The OSI Security Architecture, Security Attacks, Security Services, Security Mechanisms, A Model for Network Security, Standards. Symmetric Encryption and Message Confidentiality: Symmetric Encryption Principles, Symmetric Block Encryption Algorithms, DES, AES, Stream. 9 Hours

UNIT – II

Public-Key Cryptography and Message Authentication: Approaches to Message Authentication, Secure Hash Functions, Message Authentication Codes, Public Key Cryptography Principle, PublicKey Cryptography Algorithms, RSA, Diffie Hellman key Exchange, Digital Signatures. Key Distribution and User Authentication: Symmetric Key Distribution Using Symmetric Encryption, Kerberos, Key Distribution Using Asymmetric Encryption. 9 Hours

UNIT - III

Transport-Level Security: Web Security Issues, Secure Sockets Layer (SSL), Transport Layer Security (TLS), HTTPS, Secure Shell (SSH). Wireless Network Security: Wireless Security, Mobile Device Security, IEEE 802.11 Wireless LAN Overview, IEEE 802.11i Wireless LAN Security. Understand various security protocols and mobile security issues.(L2). 9 Hours

UNIT –IV

Electronic Mail Security: Pretty Good Privacy (PGP), S/MIME. IP Security: IP Security Overview, IP Security Policy, Encapsulating Security Payload, Combining Security Associations, Internet Key Exchange and Cryptographic Suites.

9 Hours

UNIT - V

Malicious Software: Types of Malicious Software, Propagation, Infected Content, Viruses Propagation, Vulnerability Exploit, Worms, Propagation, Social Engineering, SPAM, Trojans, Payload – System Corruption, Payload, Attack Agent, Zombie, Bots, Payload, Information Theft, Keyloggers, Phishing, Spyware, Payload, Stealthing, Backdoors, Rootkits Distributed Denial of Service Attacks.

9 Hours

Text Books:

1. Network Security Essentials, Applications and Standards, 5th edition, William Stallings, Pearson Education, 2013.

Reference Books:

1. Cryptography & Network Security, Behrouz A. Forouzan, Tata McGraw-Hill, New Delhi, 2007.
2. Network Security: Private Communication in a Public World, Kaufman, Pearson Education Asia, New Delhi, 2002.

3. Cryptography and Network Security: AtulKahate, Tata McGraw-Hill Publishing Company Limited, New Delhi, 2003.

Course Outcomes:

After completing the course student will be able to:

CO1: Understand the fundamental concepts of computer security, including the OSI security architecture, symmetric encryption, and algorithms like DES and AES, to explain secure communication principles. (L2: Understand)

CO2: Apply principles of public-key cryptography, message authentication, and key distribution techniques, including RSA and Kerberos, to secure communications and verify authenticity. (L3: Apply)

CO3: Analyse transport-level security mechanisms, such as SSL, TLS, and SSH, as well as wireless security standards like IEEE 802.11i, to address mobile and wireless network vulnerabilities. (L4: Analyse)

CO4: Evaluate email security protocols, including PGP and S/MIME, and IP security measures, such as IPsec and Internet Key Exchange, to protect sensitive information. (L5: Evaluate)

CO5: Develop strategies to identify, mitigate, and defend against malicious software attacks, including viruses, worms, Trojans, and DDoS attacks, to enhance system security. (L3: Develop)

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	2	2	0	0	3
CO2	3	3	3	3	2	0	0	3
CO3	3	3	3	3	2	0	0	3
CO4	3	3	3	3	2	0	0	3
CO5	3	3	3	2	2	0	0	3

Note: 1 - Low Correlation 2 - Medium Correlation 3 - High Correlation

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DIGITAL FORENSICS

L	T	P	S	J	C
3	0	0	0	0	3

Course Description:

This course offers an in-depth exploration into the realm of wireless and mobile technologies, focusing on the security challenges and vulnerabilities inherent in these systems, as well as the methodologies employed in digital forensics to address criminal activities involving these technologies. Students will gain insights into various wireless platforms from personal area networks to wide area networks, and delve into mobile system security frameworks, mobile forensics, and the broader field of digital forensics.

Course Objectives:

- To understand Wireless and Mobile Security Challenges
- To explore Security Frameworks for Mobile Systems
- To conduct Mobile Forensics Investigations
- To learn Principles of Digital Forensics
- To Analyse and Apply Digital Forensic Techniques.

UNIT-I

Introduction to Wireless Technologies: Overview of wireless technologies and security: Personal Area Networks, Wireless Local Area Networks, Metropolitan Area Networks, Wide Area Networks. Wireless threats, vulnerabilities and security: Wireless LANs, War Driving, War Chalking, War Flying, Common Wi-Fi security recommendations, PDA Security, Cell Phones and Security, Wireless DoS attacks, GPS Jamming, Identity theft. 9 Hours

UNIT-II

Security Framework for Mobile Systems :CIA triad in mobile Phones-Voice, SMS and Identification data interception in GSM: Introduction, practical setup and tools, implementation-Software and Hardware Mobile phone tricks: Netmonitor, GSM network service codes, mobile phone codes, catalog tricks and AT command set- SMS security issues. 9 Hours

UNIT-III

Mobile Phone Forensics : Crime and mobile phones, evidences, forensic procedures, files present in SIM card, device data, external memory dump, evidences in memory card, operators systems- Android forensics: Procedures for handling an android device, imaging android USB mass storage devices, logical and physical techniques. 9 Hours

UNIT -IV

Introduction to Digital Forensics : Digital forensics: Introduction – Evidential potential of digital devices: closed vs. open systems, evaluating digital evidence potential- Device handling: seizure issues, device identification, networked devices and contamination. 9 Hours

UNIT-V

Analysis of Digital Forensic Techniques: Digital forensics examination principles: Previewing, imaging, continuity, hashing and evidence locations- Seven element security model- developmental model of digital systems audit and logs- Evidence interpretation: Data content and context. 9 Hours

Text Books :

1. Iosif I. Androulidakis, "Mobile phone security and forensics: A practical approach", Springer publications, 2012.
2. Andrew Hoog, "Android Forensics: Investigation, Analysis and Mobile Security for Google Android", Elsevier publications, 2011.
3. Angus M. Marshall, "Digital forensics: Digital evidence in criminal investigation", John – Wiley and Sons, 2008
4. Gregory Kipper, "Wireless Crime and Forensic Investigation", Auerbach Publications.

Course Outcomes:

After completing the course student will be able to:

- **CO1:** Understand the fundamentals of wireless technologies, threats, vulnerabilities, and security recommendations to safeguard wireless communication systems. (*L2: Understand*)
- **CO2:** Apply the principles of mobile system security, including the CIA triad, GSM interception, and SMS security, to identify and mitigate mobile device vulnerabilities. (*L3: Apply*)
- **CO3:** Analyse mobile phone forensic procedures, including SIM card and memory data extraction, to collect and process digital evidence for forensic investigations. (*L4: Analyse*)
- **CO4:** Evaluate digital forensic procedures, including device seizure, identification, and contamination prevention, to ensure proper handling and assessment of digital evidence. (*L5: Evaluate*)
- **CO5:** Create a structured approach for analyzing digital forensic techniques, including imaging, hashing, and evidence interpretation, to enhance the reliability and accuracy of forensic investigations. (*L6: Develop*)

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	2	2	0	0	3
CO2	3	3	3	3	2	0	0	3
CO3	3	3	3	2	2	0	0	3
CO4	3	3	3	2	2	0	0	3
CO5	3	3	3	2	2	0	0	3

Note: 1 - Low Correlation 2 - Medium Correlation 3 - High Correlation

APPROVED IN:	
BOS : 8.6.2024	ACADEMIC COUNCIL:4.7.2024
SDG No. & Statement:	
4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.	
SDG Justification:	

The topics included in this course are designed to get acquainted with one of the skills that handle necessary mathematical orientation, programming techniques and concept based learning.

Prepared by Dr.V.Nagalakshmi
Dr.M.Srivenkatesh

Verified by Dr.M.Srivenkatesh

DIGITAL MARKETING

L	T	P	S	J	C
3	0	0	0	0	3

Course Description:

This course provides a comprehensive overview of digital marketing, exploring the various components, strategies, and tools essential for success in the digital landscape. Students will learn about the differences between digital and traditional marketing, the benefits of digital marketing, and current trends that shape the industry. The curriculum covers key digital marketing channels and techniques, including SEO (Search Engine Optimization), SEM (Search Engine Marketing), social media marketing, content marketing, email marketing, and mobile marketing.

Course Objectives:

- To understand the Fundamentals of Digital Marketing
- To develop Digital Marketing Strategies
- To master SEO and SEM Techniques
- To leverage Digital Marketing Channels
- To analyse and Optimize Digital Marketing Performance

UNIT-I

Understanding Digital Marketing: Components of Digital Marketing, Need and Scope of Digital Marketing, Benefits of Digital Marketing, Digital Marketing Platforms and Strategies, Comparison of traditional marketing and Digital marketing, Digital Marketing Trends.

Digital Marketing Plan: Elements of a Digital Marketing Plan, Marketing Plan, Executive Summary, Mission, Situational Analysis, Opportunities and Issues, Goals and Objectives, Marketing Strategy, Action Plan, Budget, Writing the Marketing Plan and Implementing the Plan. Keyword analysis. 9 Hours

UNIT-II

Marketing in the Digital Era: Keyword Planner in Google Search Console, Analysis of keyword dataset, Sample case study on Keyword dataset, Importance of Audience Segmentation, segments usage in Digital Media, Organizational Characteristics, Purchasing Characteristics, Acquisition and Retention of new customers.

Channels of Digital Marketing: Website Planning and creation tools, Search Engine Optimization, Online Advertising, PPC advertising, Google AdWords, Email Marketing, Blog Marketing, Social Media Marketing, Audio, Video and Interactive Marketing, Mobile Marketing, Migrating from Traditional Channels to Digital Channels. 9 Hours

UNIT-III

Search Engine Optimization : Introduction to On page SEO, SEO factors and tags that influence website performance, Title tag, Description tag, Header tag, Alt tag, URL structure, Robots.txt file, Sitemap, Canonicalization, Bread Crumb Structure, 301 Redirection, Custom 404 Error, Web Content Optimization, Text to Html Ratio, Website Loading Speed factors

Off page Optimization : Introduction to Off page SEO, Checking Domain Authority of website, Factors influence the high authority sites, Directory Submissions, Social Bookmarking, Forum Submission, Business Profile Creation, Article Submission, Creating backlinks, Internal, External links and Footer links, Competitor back link and data analysis. 9 Hours

UNIT- IV

Online Advertising and analysis using Web masters: Google Web Masters & Audit tools, PPC Advertising & Google AdWords, Google Ad Metrics and Performance, Payment Methods of Online Advertising – CPM (Cost-per-Thousand) and CPC (Cost-per-click), Display Ads - choosing a Display Ad Format, Landing Page and its importance, Sample Case study on online advertisement.

Google MyBusiness & Monitoring, Google Algorithms, E-Mail Marketing, SMS Marketing, Lead Generation for Business, Affiliate Marketing, Making Money through Google Adsense

9 Hours

UNIT-V

Social Media Marketing and other marketing: Influence of Social Media, Social Media Marketing, Google Analytics, Analyzing Digital Media Performance, Analyzing Website Performance, Analyzing Advertising Performance, Networking with Facebook, LinkedIn, Blogging as a social medium, Microblogging with Twitter, Social Sharing with YouTube, Social Media for Customer Reach, Acquisition and Retention. 9 Hours

Text Books:

1. B2B Digital Marketing by Michael Miller, 1e, Pearson, 2014.
2. Digital marketing by Vandana Ahuja, Oxford University Press 2015
3. Social Media Marketing by Michael R Solomon, Tracy Tuten, Pearson, 1e, 2015.
4. E-Marketing by Judy Strauss & Raymond Frost, Pearson, 2016
5. Online marketing – A customer led approach by Richard Gay, Alan Charles worth, Rita Esen,

Course Outcomes:

After completing the course student will be able to:

- **CO1:** Understand the components, strategies, and trends of digital marketing, including the creation and implementation of a comprehensive digital marketing plan. **(L2: Understand)**
- **CO2:** Apply keyword analysis, audience segmentation, and various digital marketing channels like SEO, PPC advertising, and social media to attract and retain customers. **(L3: Apply)**
- **CO3:** Analyze on-page and off-page SEO techniques, such as optimizing website performance, backlink creation, and competitor analysis, to improve search engine rankings. **(L4: Analyze)**
- **CO4:** Evaluate online advertising strategies, including Google Ad Metrics, payment methods, and landing page optimization, to design and implement effective campaigns. **(L5: Evaluate)**
- **CO5:** Develop social media marketing strategies and analyze digital media performance using tools

like Google Analytics, blogging, and social sharing platforms to enhance customer engagement and retention. (L3: Develop)

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	3	3	2	2	0	0	3
CO2	2	3	3	2	2	0	0	3
CO3	2	3	3	2	2	0	0	3
CO4	2	3	3	2	2	0	0	3
CO5	2	3	3	2	2	0	0	3

Note: 1 - Low Correlation 2 - Medium Correlation 3 - High Correlation

APPROVED IN:	
BOS : 8.6.2024	ACADEMIC COUNCIL:4.7.2024
SDG No. & Statement:	
4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.	
SDG Justification:	
The topics included in this course are designed to get acquainted with one of the skills that handle necessary mathematical orientation, programming techniques and concept based learning.	

E-COMMERCE STRATEGIES AND TECHNOLOGIES

L	T	P	S	J	C
3	0	0	0	0	3

Course Description:

This course provides an in-depth exploration of electronic commerce (e-commerce) and its profound impact on business practices and society. It covers the fundamental concepts, technologies, and business models that underpin e-commerce. Students will learn about the scope, benefits, and limitations of e-commerce, various business models, and the role of the Internet in transforming commerce. The course also delves into e-marketing, e-payment systems, e-finance, legal frameworks related to online business, and security measures necessary to protect online transactions.

Course Objectives:

- To understand the Foundations of E-Commerce.
- To explore E-Commerce Business Models: Master
- To e-Marketing and Online Strategies
- To navigate E-Payment Systems and E-Finance
- To address Legal and Security Issues in E-Commerce

UNIT -I

E – Commerce: Meaning, definition, features, functions of E-Commerce, Scope, Benefits and limitations of E-Commerce – The Internet and India – E-commerce opportunities and challenges for Industries.

Business Models for E-commerce: The Birth of Portals – E-Business Models – Business-to-Consumer (B2C) – Business-to-Business (B2B) – Consumer-to Consumer (C2C) – Consumer to-Business (C2B) – Brokerage Model – Value Chain Model – Advertising Model. 9 Hours

UNIT –II

E-marketing – Traditional Marketing Vs.E-Marketing – Impact of E-commerce on markets – Marketing issues in E-Marketing – Online Marketing – E-advertising – Internet Marketing Trends – E-Branding – Marketing Strategies. 9 Hours

E-payment Systems: Digital payment Requirements – Digital Token-based E-payment systems – Benefits to Buyers – Benefits to Sellers – Credit card as E-payment system – Mobile payments – smart card cash payment system – Micropayment system – E- Cash.

UNIT – III

E-Finance: Areas of Financing, E-Banking - Traditional Banking Vs. E-Banking – Operations in E-Banking – E-Trading – Stock Market trading – Importance and advantages of E-Trading.
Legal Framework for E-Commerce: E-Commerce Legal Framework – Rights and Obligations in the World of E-commerce – Copyrights – Defamation – Privacy – Contracts – Taxation – Signing a contract Electronically – Domain name and Registration. 9 Hours

UNIT –IV

E-Security: Security for E-commerce – Security Design – Analysing risk – E-Banks and Security – safety of E-Commerce – Online Shopping with confidence – Firewalls for system Integrity – Virus Protection and Protection from intruders. 9 Hours

UNIT –V

Mobile Commerce: Challenges of E-commerce – Global Mobile E-Commerce – Secure Mobile Commerce – Secured Payments through Mobile – First Mobile Commerce Service. 9 Hours

TEXT BOOK: Joseph P. T., E - Commerce – An Indian Perspective

REFERENCE: Jaiswal S., E-Commerce

Mohammad Mahmoudi Maymand, E-Commerce

Murthy C.S.V., E-Commerce - Concepts, Models and Strategies

Course Outcomes:

After completing the course student will be able to:

- **CO1:** Understand the features, functions, and business models of E-commerce, including B2C, B2B, C2C, and C2B, to identify opportunities and challenges in the digital market. (**L2: Understand**)
- **CO2:** Apply E-marketing strategies, including online advertising and branding, and evaluate digital payment systems like credit cards, mobile payments, and e-cash to enhance customer engagement and transaction efficiency. (**L3: Apply**)
- **CO3:** Analyse the impact of E-banking, E-trading, and legal frameworks on E-commerce operations, including rights, obligations, and electronic contracts, to ensure compliance and efficiency. (**L4: Analyse**)
- **CO4:** Evaluate E-commerce security mechanisms, including firewalls, virus protection, and risk management, to protect systems from cyber threats and ensure secure transactions. (**L5: Evaluate**)
- **CO5:** Develop secure mobile commerce solutions by addressing global challenges and integrating secure payment systems for effective mobile-based E-commerce services. (**L3: Develop**)

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	2	2	0	0	3
CO2	2	3	3	3	2	0	0	3
CO3	2	3	3	3	2	0	0	3
CO4	2	3	3	2	2	0	0	3
CO5	3	3	3	3	3	0	0	3

Note: 1 - Low Correlation 2 - Medium Correlation 3 - High Correlation

APPROVED IN:	
BOS : 8.6.2024	ACADEMIC COUNCIL:4.7.2024
SDG No. & Statement:	
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SDG Justification:	
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Prepared by Dr.G,Ramash Naidu

Verified by Dr.M.Srivenkatesh

SOCIAL NETWORK ANALYSIS

L	T	P	S	J	C
3	0	0	0	0	3

Course Description:

This course offers a comprehensive exploration of social networks and social media from a data analysis perspective, covering the structural properties, dynamics, and impacts of networks in various contexts. Students will learn about the theory and applications of social network analysis (SNA), including the use of graphs to represent and analyze social structures. The curriculum integrates a blend of theoretical foundations with practical exercises involving network measures, centrality types, network growth models, link analysis algorithms, and community detection methods.

Course Objectives:

- To understand the Fundamentals of Social Networks
- To analyse Network Structures
- To evaluate Centrality and Influence
- To explore Network Models and Community Detection
- To apply Theories to Real-World Data

UNIT -I

Introduction: Defining Social Networks, Social Media; Social Network vs Social Media, Social Network Analysis-Introduction, Applications, Small World Phenomenon-Six degrees of separation- Milgram Experiment, Use case of Milgram experiment-Small World phenomenon in the current era of Facebook.

Graph Essentials: Graph Basics, Types of Graphs, Special Graphs in Social Networks. Introduction to Networks: Definition, Types of networks, Node vs Link-centric view, Other types of network views- local, temporal, and generalized, Three levels of Social Network Analysis.

9 Hours

UNIT -II

Network Measures: Network basics-Degree, Degree distribution, Paths, Clustering coefficients, connected components.

Node centrality Types- Degree, Closeness, Betweenness, Eigen, Katz, PageRank, Group; Hub and Authority; Assortativity, Similarity, Degeneracy, Transitivity and Reciprocity, Social Balance Theory, Similarity in Networks.

9 Hours

UNIT -III

Network Growth Models- Properties of Real-World Networks, Random Network Model, Ring lattice Network model, Watts-Strogatz, Preferential Attachment Model, Small World Model, Price's Model, Aging in Preferential Attachment.

9 Hours

UNIT -IV

Link Analysis- Strong and Weak Ties, Link Analysis Algorithm, PageRank-Matrix Manipulation and Convergence, DivRank, SimRank, PathSIM

Community Structure in Networks- Types of Communities, Community Detection Models, Community Detection- Disjoint, Overlapping, Local.

9 Hours

Unit-V:

Information Diffusion in Social Media- Herd Behaviour, Epidemics- SI, SIS, SIR, SIRS models for information diffusion, Cascade models- Probabilistic, Independent; Cascade Prediction.

Anomaly Detection in Networks- Outliers vs Network-based anomalies, Anomaly detection in static vs dynamic networks. Brief Introduction to Collusion in Online Social Networks.

9 Hours

Textbooks:

1. Tanmoy Chakraborty, Social Network Analysis, Wiley India Pvt Ltd., 2021, ISBN-978-93-5424-783-5.
2. Reza Zafarani, Mohammad Ali Abbasi, Huan Liu, Social Media Mining – An Introduction, Cambridge University Press, 2014.

Reference Books:

1. Hansen, Derek, Ben Shneiderman, Marc Smith., Analyzing Social Media Networks with NodeXL: Insights from a Connected World, Morgan Kaufmann, 2011.
2. Krishna Raj P.M, Ankit Mohan, K.G. Srinivasa, Practical Social Network Analysis with Python, Springer, Computer Communications and Networks, 2018, ISBN 978-3-319-96745-5.
3. Iribarren, José Luis, and Esteban Moro. "Information diffusion epidemics in social networks." *arXiv preprint arXiv:0706.0641* (2007).
4. David Easley, Jon Kleinberg, Networks, Crowds, and Markets: Reasoning About a Highly Connected World, Cambridge University Press, 2010.
5. Sergey Edunov, Smriti Bhagat, Moira Burke, Carlos Diuk, Ismail Onur Filiz, Three and a half degrees of separation, Meta, 2016, <https://research.facebook.com/blog/2016/2/three-and-a-half-degrees-of-separation/>

Course Outcomes:

After completing the course student will be able to:

CO1: Understand the components, strategies, and trends of digital marketing, including the creation and implementation of a comprehensive digital marketing plan. *(L2: Understand)*

CO2: Apply keyword analysis, audience segmentation, and various digital marketing channels like SEO, PPC advertising, and social media to attract and retain customers. *(L3: Apply)*

CO3: Analyse on-page and off-page SEO techniques, such as optimizing website performance, backlink creation, and competitor analysis, to improve search engine rankings. *(L4: Analyse)*

CO4: Evaluate online advertising strategies, including Google Ad Metrics, payment methods, and landing page optimization, to design and implement effective campaigns. *(L5: Evaluate)*

CO5: Develop social media marketing strategies and analyse digital media performance using tools like Google Analytics, blogging, and social sharing platforms to enhance customer engagement and retention. *(L6: Develop)*

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	3	3	3	3	3
CO2	2	2	2	2	2	2	2	2
CO3	1	1	1	1	1	1	1	1
CO4	-	-	-	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-

Note: 1 - Low Correlation 2 - Medium Correlation 3 - High Correlation

APPROVED IN:	
BOS : 8.6.2024	ACADEMIC COUNCIL:4.7.2024
SDG No. & Statement:	
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COMPILER DESIGN

L	T	P	S	J	C
3	0	0	0	0	3

Course Description:

This course delves into the fundamentals of compiler design, introducing the essential components and processes involved in translating source code written in a high-level programming language into executable machine code. Students will explore the architectural structure of compilers, including phases such as lexical analysis, syntax analysis, syntax-directed translation, intermediate code generation, code optimization, and code generation. The course incorporates practical tools and methods used in compiler construction, such as LEX/FLEX for lexical analysis and YACC for parser generation.

Course Objectives:

- To understand Compiler Structure and Functionality
- To master Lexical Analysis
- To develop Skills in Syntax Analysis
- To implement Syntax-Directed Translation and Intermediate Code Generation
- To optimize and Generate Target Code

UNIT-I

Introduction & Lexical Analysis

Introduction: The Structure of Compiler, The Science of Building a Compiler in Bootstrapping and Cross compiler.

Lexical Analysis: The role of the Lexical analyzer, Input Buffering, Specification of Tokens, Recognition of Tokens, The Lexical Analyzer Generator (LEX/FLEX). 9 Hours

UNIT-II

Syntax Analysis (Part-I)

Syntax Analysis (Part-I): Introduction, Context-Free Grammars, Top-Down parsing: Brute force Parsing, Recursive Descent Parsing, Non-recursive Predictive Parsing, Error Recovery in Predictive Parsing, Bottom- Up parsing - Shift Reduce Parsing, Operator-Precedence Parsing, Error Recovery in Operator Precedence Parsing. 9 Hours

UNIT-III

Syntax Analysis (Part-II)

Introduction to LR Parsing: Simple LR Parser, More Powerful LR Parsers (CLR&LALR), Using Ambiguous grammars, Error Recovery in LR Parsers, Parser Generator (YACC). 9 Hours

UNIT-IV

Syntax Directed Translation & Intermediate Code

Generation

Syntax Directed Translation: Syntax Directed Definitions, Types of attributes, Evaluation Orders for SDD's, Applications of Syntax-Directed Translation.

Intermediate Code Generation: Three Address codes, Types & Declarations, Translation of Arithmetic Expressions, Back patching for Boolean Expressions, and Flow of Control. 9 Hours

UNIT-V

Code Optimization & Code Generation

Code Optimization: The Principal Sources of Optimization, Basic blocks and Flow Graphs, Optimization of Basic Blocks, Introduction to Data-Flow Analysis: Live Variable Analysis

Code Generation: Issues in designing a code Generator, The Target Language, A Simple Code Generator, Peephole Optimization, Register allocation, and Assignment. 9 Hours

TextBooks:

1. Alfred.V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey. D. Ullman, Compilers Principles, Techniques and Tools, 2/e, Pearson Education, 2008.

References:

1. Alfred.V. Aho, J.D.Ullman, Principles of compiler design, Narosa Publications, 2002

2. John R. Levine, Tony Mason, Doug Brown, Lex &yacc, O'reilly ,2/e, 1992

3. Keith Cooper, Linda Torczon, Engineering a compiler, Morgan Kaufmann, 2/e, 2011

4. Samuel, M. P., Philip, P. K., 1997, "Powder metallurgy tool electrodes for electric discharge machining", International journal of machine tools and manufacture, 37, 1625–33.

Course Outcomes:

After completing the course student will be able to:

CO1: Understand the features, functions, and business models of E-commerce, including B2C, B2B, C2C, and C2B, to identify opportunities and challenges in the digital market. *(L2: Understand)*

CO2: Apply E-marketing strategies, including online advertising and branding, and evaluate digital payment systems like credit cards, mobile payments, and e-cash to enhance customer engagement and transaction efficiency. *(L3: Apply)*

CO3: Analyse the impact of E-banking, E-trading, and legal frameworks on E-commerce operations, including rights, obligations, and electronic contracts, to ensure compliance and efficiency. *(L4: Analyse)*

CO4: Evaluate E-commerce security mechanisms, including firewalls, virus protection, and risk management, to protect systems from cyber threats and ensure secure transactions. *(L5: Evaluate)*

CO5: Develop secure mobile commerce solutions by addressing global challenges and integrating secure payment systems for effective mobile-based E-commerce services. *(L3: Develop)*

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	3	2	0	0	3
CO2	3	3	3	3	2	0	0	3
CO3	3	3	3	3	2	0	0	3
CO4	3	3	3	3	2	0	0	3
CO5	3	3	3	3	2	0	0	3

Note: 1 - Low Correlation 2 - Medium Correlation 3 - High Correlation

APPROVED IN:	
BOS : 8.6.2024	ACADEMIC COUNCIL:4.7.2024
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SDG Justification:	
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MACHINE LEARNING LAB

L	T	P	S	J	C
0	0	2	0	0	1

Course Description:

This laboratory course provides hands-on experience in machine learning using Python, focusing on practical applications of data manipulation, exploratory data analysis, data visualization, and predictive modelling. Throughout the course, students will work with popular Python libraries such as Pandas for data handling and analysis, Matplotlib and Seaborn for data visualization, and Scikit-learn for implementing machine learning algorithms. The lab exercises are designed to reinforce the theoretical concepts covered in machine learning courses by applying them to real-world datasets and scenarios.

Course Objectives:

- To master Data Manipulation and Analysis Using Pandas
- To conduct Exploratory Data Analysis (EDA)
- To develop Skills in Data Visualization
- To implement Machine Learning Models
- To evaluate Model Performance

Lab Exercises

1. Creating a Data Frame in Pandas from csv files.
2. Importing Data with Pandas – adding columns to the data frame.
3. Handling Missing Data- drop, fill, aggregate functions.
4. Indexing Data Frames with Pandas, Indexing Using Labels in Pandas.
5. Exploratory Data Analysis with Pandas- for both one dimensional and two dimensional data (series or data frames) - describe, group data, ANOVA, correlation and correlation methods, rank.
6. Calculating Mean, Trimmed Mean, Weighted Mean, Median,
7. Plotting using pandas- Exploratory analysis based on the plots.
8. Data Visualization with different charts in python.
9. A weather prediction model that predicts if there'll be rain or not in a particular day with decision tree regression concept.
10. A Python script to create a confusion matrix on a predicted model.
11. Write a program demonstrates how to use K-Nearest Neighbors (KNN), Decision Trees, Random Forest, and Support Vector Machines (SVM) for classification
12. Write a program demonstrates linear regression, multiple linear regression, and polynomial regression using synthetic data.
13. This program demonstrates K-Means and hierarchical clustering using synthetic data.

Text Books:

1. Machine Learning by Subramanian, Chandra Mouli, Amit Kumar Das , Saikant Dutt, Pearson Publications, I edition, 2018.

2. Machine Learning by Tom Mitchell, McGraw Hill,

Course Outcomes:

After completing the course student will be able to:

- **CO1:** Understand the fundamental concepts of social networks, social media, and graph essentials, including network views and levels of social network analysis, to explore real-world social interactions.
- **CO2:** Apply network measures such as centrality, clustering coefficients, assortativity, and social balance theory to analyse the structure and dynamics of social networks. (**L3: Apply**)
- **CO3:** Analyse network growth models, including random networks, preferential attachment, and small-world models, to study the properties of real-world social networks. (**L4: Analyse**)
- **CO4:** Evaluate link analysis algorithms like PageRank, SimRank, and community detection models to understand community structures and relationships within networks. (**L5: Evaluate**)
- **CO5:** Develop models for information diffusion and anomaly detection in social networks to predict cascading behaviors, identify outliers, and detect collusion in online platforms. (**L3: Develop**)

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	2	3	2	0	0	3
CO2	3	3	2	3	2	0	0	3
CO3	3	3	2	3	2	0	0	3
CO4	3	3	3	3	2	0	0	3
CO5	3	3	3	3	2	0	0	3

Note: 1 - Low Correlation 2 - Medium Correlation 3 - High Correlation

APPROVED IN:	
BOS : 8.6.2024	ACADEMIC COUNCIL:4.7.2024
SDG No. & Statement:	
4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.	
SDG Justification:	
The topics included in this course are designed to get acquainted with one of the skills that handle necessary mathematical orientation, programming techniques and concept based learning.	

Prepared by Dr.M.Srivenkatesh
Mrs Shahid Shaik

Verified by Dr.M.Srivenkatesh

SERVER-SIDE PROGRAMMING WITH JAVA LAB

L	T	P	S	J	C
0	0	2	0	0	1

Course Description:

This laboratory course is designed for students to apply Java technologies in full stack development, covering both front-end and back-end components. Through a series of hands-on lab exercises, students will gain practical experience in using Java Servlets, JavaServer Pages (JSP), Java Persistence API (JPA), Hibernate, and Spring frameworks. The course focuses on building dynamic web applications, implementing server-side programming, managing sessions, interacting with databases, and developing secure APIs.

Course Objectives:

- To understand and apply Java Servlet technology.
- To develop Dynamic Web Applications
- To implement Database Interactions
- To utilize ORM Techniques
- To leverage Spring Framework

Lab Exercise Programs

1. Write A JAVA Servlet Program To Implement And Demonstrate Get() And Post Methods (Using HTTP Servlet Class).
2. Write A JAVA Servlet Program To Implement Request dispatcher Object (Use Include() And Forward() Methods).
3. Write A JAVA Servlet Program To Implement Sessions (Using HTTP Session Interface).
4. Write A JSP Program Which Uses Jsp:Include & Jsp:Forward Action To Display A Web Page.
5. Write A JSP Program To Implement All The Attributes Of Page Directive Tag.
6. write a JSP program to upload image file into server location?
7. Write a JSP which does the following job: Insert the details of the 3 or 4 users who register with the web site by using registration form. Authenticate the user when he/she submits the login form using the user name and password from the database
8. Create tables in the database which contain the details of items (books in our case like Book name, Price, Quantity, Amount)) of each category. Modify your catalogue page in such a way that you should connect to the database and extract data from the tables and display them in the catalogue page using JDBC.
9. Write a java JSP program to connect to that database and execute various SQL queries.
10. Develop a JPA Application to demonstrate use of ORM associations
11. Develop a Hibernate application to store and retrieve employee details in Database
12. Create a simple application that uses Spring for dependency injection.
13. Create a simple REST API using Spring Boot.
14. Demonstrate transaction management with Spring boot

Text Books :

- 1.Head First Servlets and JSP: Passing the Sun Certified Web Component Developer Exam
Bryan Basham, Kathy Sierra, and Bert Bates, O'Reilly Media, 2nd Edition (2008)
- 2.Full Stack Java Development with Spring MVC, Hibernate, jQuery, and Bootstrap Mayur Ramgir,Kindle Edition,

Course Outcomes:

After completing the course student will be able to:

- **CO1:** Understand the core concepts of Java Servlets and JSP, including the use of HTTP methods, request dispatchers, and session management, to develop dynamic web applications. **(L2: Understand)**
- **CO2:** Apply JSP directives, file upload techniques, and database connectivity to create interactive and data-driven web pages. **(L3: Apply)**
- **CO3:** Develop JDBC and ORM frameworks like JPA and Hibernate to implement efficient data storage and retrieval systems. **(L6: Develop)**
- **CO4:** Apply dependency injection mechanisms in Spring and REST API development using Spring Boot to design modular and scalable applications. **(L3: Apply)**
- **CO5:** Develop robust enterprise-level applications by demonstrating transaction management in Spring Boot for ensuring data consistency and reliability. **(L3: Develop)**

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	3	2	1	0	3
CO2	3	3	3	3	2	1	0	3
CO3	3	3	3	3	2	1	0	3
CO4	3	3	3	3	2	1	0	3
CO5	3	3	3	3	2	1	0	3

Note: 1 - Low Correlation 2 - Medium Correlation 3 - High Correlation

APPROVED IN:	
BOS : 8.6.2024	ACADEMIC COUNCIL:4.7.2024
SDG No. & Statement:	
4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.	
SDG Justification:	

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Prepared by Mr.B.Satyanarayana

Verified by Dr.M.Srivenkatesh

SERVER-SIDE PROGRAMMING WITH PYTHON LAB

L	T	P	S	J	C
0	0	2	0	0	1

Course Description:

This laboratory course is designed to provide hands-on experience in full stack web development using Python, with a particular focus on Flask and Django frameworks. Students will engage in building functional web applications that showcase their ability to integrate both front-end and back-end technologies effectively. The course covers a broad range of activities from setting up simple web servers to developing complex RESTful APIs and deploying web applications to production environments like Heroku.

Course Objectives:

- To develop Web Applications with Flask
- To implement Database Integration
- To construct RESTful APIs
- To master Django's MVT Architecture
- To deploy Applications to Production

Lab Exercise Programs

1. Write a Python script that makes an HTTP GET request to a specified URL and processes the JSON response to extract and print specific information.
2. Write a program for developing a simple Flask application with message.
3. Write a Flask application with basic routing that renders different HTML templates for different URLs.
4. Develop a Flask application that includes a form for user input. The application should process the input and display the result on a new page. Implement sessions to maintain user state.
5. Develop a larger Flask application using blueprints. Create at least two blueprints representing different sections of the application.
6. Develop a simple Flask application that connects to a SQLite database using SQLAlchemy.
7. Create a model for a simple entity (e.g., `User`) and perform CRUD operations through the Flask application.

8. Build a RESTful API using Flask-RESTful that supports CRUD operations on a resource
(e.g., `Product`). Implement endpoints for creating, reading, updating, and deleting resources.
9. Create a simple Django application that renders a "Hello, World!" message using the MVT pattern.
10. Create Django models for a simple entity (e.g., `Book`) with fields such as title, author, and publication date. Perform migrations and use the Django ORM to perform CRUD operations on the models.
11. Create views and class-based views to handle HTTP GET and POST requests in a Django application. Develop a form to collect user data and handle form submissions.
12. Demonstrate Django signals to perform specific actions when certain events occur
(e.g., sending an email when a new user registers).
13. Develop and deploy a Django application on Heroku environment .

Text Books :

1. Flask Web Development: Developing Web Applications with Python by Miguel Grinberg
Python Web Programming with Flask by Fabrizio Romano, O'Reilly Media, 2nd Edition , 2018
2. Django for Beginners: Build websites with Python and Django by William S. Vincent,
3rd Edition (2020)
3. Two Scoops of Django 3.x: Best Practices for the Django Web Framework by Audrey Roy
Greenfeld and Daniel Roy Greenfeld , Two Scoops Press, 1st Edition , 2020
4. Django 3 By Example by Antonio Melé, Packt Publishing, 3rd Edition , 2020

Course Outcomes:

After completing the course student will be able to:

- **CO1:** Understand the fundamentals of building web applications using Flask and Django, including routing, templates, and the MVT pattern, to create dynamic web pages. (*L2: Understand*)
- **CO2:** Apply form handling, session management, and CRUD operations using Flask and Django to develop user-interactive and database-driven web applications. (*L3: Apply*)
- **CO3:** Analyse the use of advanced Flask and Django features, including blueprints, RESTful APIs, and Django ORM, to structure scalable and maintainable web applications. (*L4: Analyse*)
- **CO4:** Evaluate the integration of event-driven features like Django signals and deploy web applications on cloud platforms like Heroku to enhance functionality and accessibility. (*L5: Evaluate*)
- **CO5:** Develop robust and efficient web solutions by combining Flask and Django frameworks with databases, RESTful APIs, and deployment strategies for real-world applications. (*L3: Develop*)

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	3	2	0	0	3
CO2	3	3	3	3	2	0	0	3
CO3	3	3	3	3	2	0	0	3
CO4	3	3	3	3	2	0	0	3
CO5	3	3	3	3	2	0	0	3

Note: 1 - Low Correlation 2 - Medium Correlation 3 - High Correlation

APPROVED IN:	
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Prepared by Dr.MSrivenkatesh
Mr. Shahid Shaik

Verified by Dr.M.Srivenkatesh

BUSINESS INTELLIGENCE LAB USING POWER BI

L	T	P	S	J	C
0	0	2	0	0	1

Course Description:

This laboratory course is designed to provide hands-on experience with Power BI, a leading business intelligence tool, to create dynamic visualizations and business intelligence reports. Students will learn to import data from various sources, perform data transformation and loading (ETL) processes, and develop interactive visualizations and dashboards that make data-driven insights accessible and actionable.

Course Objectives:

- Master Data Importation and Transformation
- Develop Skills in ETL Processes
- Create Interactive Visual Reports
- Design and Implement Dynamic Dashboards
- Apply DAX for Advanced Data Modelling

Lab Exercise Programs

1. Import the data from different data sources (xls ,OLE DB, CSV other formats) and load in the target system
2. Perform the Extraction Transformation and Loading (ETL) process to construct the database in the Power BI
3. Create an interactive chart showing Units in Stock by Product and Total Sales by Year
4. Create a stacked Area Chart using a sample diabetic patients data set to show increase in no of patients by year and month
5. Create an Interactive report using a sample data set to show Filled Map for Geographical Region
6. Create an Interactive report using a sample data set to show Bar Chart for Age Distribution
7. Create an Interactive report using a sample data set to show Pie Chart for Gender Classification
8. Create an Interactive report using a sample data set to show Tree Map for Job Classification
9. Create a dynamic report using a sample data set to show reflections in other charts.
10. Create a dynamic dashboard for diabetic patients increase with respect to age , gender and geographical area
11. Create a dynamic dashboard for a sample dataset to visualize the increases of customers based on age, c
12. Company wise and geographical area

Text Books :

1. Mastering Microsoft Power BI: Expert techniques for effective data analytics and business intelligence, Brett Powell,; Packt Publishing, 2nd Edition, 2020
2. Applied Microsoft Power BI (2nd Edition): Bring your data to life!, Authors: Teo Lachev
Publisher: Prologika Press, 2nd Edition, 2021
3. Analyzing Data with Power BI and Power Pivot for Excel", Alberto Ferrari, Marco Russo
Microsoft Press, 1st Edition ,2017

Course Outcomes:

After completing the course student will be able to:

- **CO1:** Understand the process of importing and integrating data from various sources, including Excel, OLE DB, and CSV, into Power BI for analysis and reporting. (*L2: Understand*)
- **CO2:** Apply the ETL (Extract, Transform, Load) process in Power BI to construct and transform databases for effective data visualization. (*L3: Apply*)
- **CO3:** Analyse data trends using interactive visualizations such as area charts, bar charts, pie charts, and tree maps to provide insights into patterns like stock levels, sales, and demographic distributions. (*L4: Analyse*)
- **CO4:** Evaluate dynamic reporting techniques, including reflections and interactivity across charts and maps, to enhance user engagement and decision-making. (*L5: Evaluate*)
- **CO5:** Develop dynamic dashboards to visualize complex relationships between variables, such as patient demographics and sales trends, and effectively communicate key business insights. (*L3: Develop*)

CO-PO Mapping table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	3	2	0	0	3
CO2	3	3	3	3	2	0	0	3
CO3	3	3	3	3	2	0	0	3
CO4	3	3	3	3	2	0	0	3
CO5	3	3	3	3	2	0	0	3

Note: 1 - Low Correlation 2 - Medium Correlation 3 - High Correlation

APPROVED IN:	
BOS : 8.6.2024	ACADEMIC COUNCIL:4.7.2024
SDG No. & Statement:	
4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.	
SDG Justification:	
The topics included in this course are designed to get acquainted with one of the skills that handle necessary mathematical orientation, programming techniques and concept based learning.	

Prepared by Mr. Shahid Shaik

Verified by Dr.M.Srivenkatesh

AGILE SOFTWARE DEVELOPMENT LAB

L	T	P	S	J	C
0	0	2	0	0	1

Course Description:

This laboratory course is designed to provide practical, hands-on experience with various Agile software development methodologies, including Scrum, Extreme Programming (XP), Feature-Driven Development (FDD), and Test-Driven Development (TDD). Students will engage in a series of exercises that simulate real-world software development projects to understand and apply the principles and practices of Agile development.

Course Objectives:

- Familiarize with Agile Tools and Environments
- Understand and Compare Agile Methodologies
- Apply XP and TDD Practices
- Simulate Scrum Processes
- Integrate and Implement Agile Modelling and FDD

Lab Exercise Programs

Lab 1: Agile Practices and Tools

Objective: Familiarize students with Agile tools.

Exercise: Set up and use Jira or Trello to create a project. Define epics, stories, and tasks using an example project of your choice. Explore features like backlogs, sprints, and boards.

Deliverable: A screenshot of the project setup and a brief report on the tool's features.

Lab 2: Introduction to Agile Methods

Program :Agile Methodology Comparison

Objective: Compare and contrast different Agile methodologies.

Exercise: Research and create presentations or a comparative analysis document on various Agile methodologies mentioned in the unit (XP, SCRUM, DSDM, FDD, TDD). Highlight key practices, benefits, and challenges of each.

Deliverable: Presentation or document detailing the findings.

Lab 3: Extreme Programming (XP)

Objective: Understand and apply XP practices.

Exercise: Conduct a pair programming session. Implement a simple application (e.g., a calculator or a todo list) where one student writes the test and the other writes the code to pass the test.

Deliverable: Code repository (e.g., GitHub) with commit history showing pair programming practice.

Lab 4: Extreme Programming (XP)

Program : XP Simulation

Objective: Simulate an XP environment to understand its dynamics.

Exercise: Organize a coding dojo session where students practice XP's 12 practices.

Deliverable: Code repository and a reflection on the experience and effectiveness of XP practices.

Lab 5: Test-Driven Development (TDD)

Objective: Practice TDD by developing features starting with tests.

Exercise: Develop a small software feature using TDD. Start with writing tests for the requirements

and then write the code to pass the tests.

Deliverable: Code repository with test cases and the final code that passes all tests.

Lab 6: Agile Modeling and XP

Program: Agile Modeling Workshop

Objective: Apply Agile modeling techniques in a real project.

Exercise: Create a simple system design using Agile modeling techniques. Integrate these models into an existing XP project to demonstrate the fit and benefits of Agile modeling.

- **Deliverable:** A set of Agile models (like user story maps, UML diagrams) and a brief report on how these models improved the project planning and implementation phases.

Lab 7: Scrum Simulation

Objective: Simulate a Scrum sprint to understand Scrum processes.

Exercise: Perform a two-week sprint simulation where students perform roles such as Scrum Master, Product Owner, and Team Members. Use daily standups, sprint planning, sprint review, and retrospective.

Deliverable: Sprint report including burndown chart, user stories completed, and retrospective findings.

Lab 8: Scrum Simulation

Program: Scrum Role Play

Objective: Experience the roles within a Scrum team.

Exercise: Conduct a role-playing session where students take on the roles of Product Owner, Scrum Master, and Development Team members. Tackle a mock project from product backlog creation to sprint review.

Deliverable: Role play script, product backlog, sprint backlog, and a sprint review report.

Lab 9: Feature-Driven Development (FDD) and Incremental Software Development

Objective: Implement a feature using FDD principles.

Exercise: Plan and develop a feature using the FDD methodology. Focus on building by feature and track progress in a visual way.

Deliverable: Documentation of the planning and development process, including diagrams and

final code.

Lab 10: Feature-Driven Development and Test-Driven Development

Program: FDD and TDD Integration Project

Objective: Integrate FDD planning with TDD execution.

Exercise: Plan a small project using FDD, breaking down the project into features. Implement one of these features using TDD practices, focusing on writing tests first and then coding to fulfill the tests.

Deliverable: Document outlining the FDD plan, tests written, code developed, and how TDD influenced feature implementation.

Text Books:

1. John Hunt, Agile Software Construction, 1st Edition, Springer, 2005
2. Craig Larman, Agile and Iterative Development: A Manager's Guide, Addison-Wesley, Pearson Education – 2004.
3. Pearson, Robert C. Martin, Juli, James Shore, Chromatic 2013, The Art of Agile Development, O'Reilly Media.
4. Elisabeth Hendrickson, Agile Testing, Quality Tree Software Inc 2008.

Course Outcomes:

After completing the course student will be able to:

- **CO1:** Understand the fundamental principles of Agile methodologies and tools such as Jira or Trello to create and manage Agile projects. (*L2: Understand*)
- **CO2:** Apply Extreme Programming (XP) practices such as pair programming, test-first coding, and Agile modeling to develop and manage software projects effectively. (*L3: Apply*)
- **CO3:** Analyse the dynamics of Agile practices like Scrum and Feature-Driven Development (FDD) by simulating sprints and implementing feature-based development. (*L4: Analyse*)
- **CO4:** Evaluate the integration of Agile methodologies like FDD and Test-Driven Development (TDD) to enhance software quality and project planning. (*L5: Evaluate*)
- **CO5:** Develop comprehensive Agile project deliverables by integrating Agile tools, techniques, and methodologies such as TDD, FDD, and Scrum to solve real-world software development problems. (*L3: Develop*)

CO-PO Mapping table

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CO1	3	2	3	3	3	0	0	3
CO2	3	3	3	3	3	0	0	3
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CO5	3	3	3	3	3	0	0	3

Note: 1 - Low Correlation 2 - Medium Correlation 3 - High Correlation

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CLOUD COMPUTING LAB

L	T	P	S	J	C
0	0	2	0	0	1

Course Description:

This laboratory course is designed to provide students with practical, hands-on experience in cloud computing across multiple platforms including Microsoft Azure, Amazon Web Services (AWS), OpenAI, and Google Cloud Platform. Through a series of guided exercises, students will learn to configure and deploy virtual machines, manage cloud storage, develop serverless applications, and implement AI-powered applications using cloud APIs.

Course Objectives:

- To gain Proficiency with Major Cloud Platforms
- To master Virtual Machine Deployment and Management
- To deploy and Manage Web and Serverless Applications
- To implement Cloud Storage Solutions
- To develop AI Applications with Cloud APIs

Microsoft Azure Lab Exercises

1. Creating a Virtual Machine:

- Objective: Create and configure a virtual machine in Azure.
- Steps:
 - Create a new virtual machine.
 - Configure the operating system and network settings.
 - Connect to the virtual machine via Remote Desktop.
 - Install and configure necessary software.

2. Azure App Services:

- Objective: Deploy a web application using Azure App Services.
- Steps:
 - Create a new web app in Azure.
 - Deploy a sample application using Azure App Service.
 - Configure application settings and scaling options.
 - Monitor the app's performance and usage.

3. Azure Storage:

- Objective: Utilize Azure Storage for data storage.
- Steps:
 - Create a storage account.
 - Use Blob Storage to upload and manage files.
 - Configure access control and generate SAS tokens.
 - Implement a sample application to interact with Azure Storage.

AWS Lab Exercises

1. Creating an EC2 Instance:

- Objective: Launch and configure an EC2 instance.
- Steps:
 - Launch an EC2 instance using the AWS Management Console.
 - Configure security groups, key pairs, and network settings.
 - Connect to the instance using SSH.
 - Install and configure necessary software.

2. Deploying a Serverless Application with AWS Lambda:

- Objective: Create a serverless application using AWS Lambda.
- Steps:
 - Create a Lambda function.
 - Configure triggers and permissions.
 - Deploy and test the Lambda function.
 - Integrate Lambda with other AWS services like S3 and DynamoDB.

3. AWS S3 Bucket Management:

- Objective: Manage data using AWS S3.
- Steps:
 - Create an S3 bucket.
 - Upload, download, and manage objects in the bucket.
 - Configure bucket policies and permissions.
 - Implement versioning and lifecycle policies.

OpenAI Tools Lab Exercises

1. Chatbot Development using OpenAI GPT:

- Objective: Create a chatbot using OpenAI GPT API.
- Steps:
 - Set up the OpenAI API environment.
 - Create a basic chatbot application.
 - Implement conversation flows and context handling.
 - Test and deploy the chatbot.

2. Text Generation with OpenAI GPT:

- Objective: Generate text using OpenAI GPT.
- Steps:
 - Set up the OpenAI API environment.
 - Create a script to generate text based on prompts.
 - Experiment with different prompt designs and parameters.
 - Analyze the generated text and refine the prompts.

3. Sentiment Analysis using OpenAI:

- Objective: Perform sentiment analysis using OpenAI GPT.
- Steps:
 - Set up the OpenAI API environment.
 - Create a script to analyze sentiment in given text inputs.
 - Test the script with various text samples.
 - Integrate sentiment analysis into a larger application.

Google Cloud Lab Exercises

1. Creating a Compute Engine Instance:

- Objective: Create and configure a Google Cloud Compute Engine instance.
- Steps:
 - Create a new Compute Engine instance.
 - Configure the machine type, disk, and network settings.
 - Connect to the instance using SSH.
 - Install and configure necessary software.

2. Google Cloud Storage:

- Objective: Utilize Google Cloud Storage for data storage.
- Steps:
 - Create a Cloud Storage bucket.
 - Upload and manage objects in the bucket.
 - Configure access control and permissions.
 - Implement a sample application to interact with Cloud Storage.

3. Deploying Applications using Google App Engine:

- Objective: Deploy a web application using Google App Engine.
- Steps:
 - Create a new App Engine project.
 - Deploy a sample web application.
 - Configure application settings and scaling options.
 - Monitor the app's performance and usage.

Course Outcomes:

After completing the course student will be able to:

- **CO1:** Understand the fundamental concepts of cloud computing platforms such as Microsoft Azure, AWS, Google Cloud, and OpenAI tools to deploy and manage applications and services. (L2: Understand)
- **CO2:** Apply cloud-based services for creating and configuring virtual machines, deploying applications, and managing storage solutions. (L3: Apply)
- **CO3:** Analyse the integration of serverless architectures, APIs, and advanced cloud tools to optimize application deployment and functionality. (L4: Analyse)
- **CO4:** Evaluate the performance and scalability of cloud applications through monitoring tools, configuration settings, and resource management. (L5: Evaluate)
- **CO5:** Develop interactive and scalable cloud-based solutions, such as chatbots, sentiment analysis applications, and dynamic storage systems, by leveraging cloud-native tools and APIs. (L3: Develop)

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INTERNSHIP /INDUSTRIAL TRAINING AND SEMINAR

L	T	P	S	J	C
0	0	0	0	0	2

Course Duration: 4 Weeks

Course Description: This introductory internship program is designed to bridge the gap between academic theories and real-world industry practices. Students will spend a four weeks at an industry site, gaining first-hand experience under the supervision of industry experts. This exposure prepares them for future engagements such as more advanced internships or full-time positions.

Course Educational Objectives:

- Gain practical exposure to industry standards and teamwork.
- Identify essential skills required for effective participation in real-time industry projects.
- Develop an understanding of the broader implications of engineering solutions, including their global, economic, environmental, and societal impacts.
- Acquire the foundational technical knowledge necessary for advancing to more complex internships.
- Enhance problem-solving skills by addressing real-world challenges, encouraging innovative and critical thinking approaches.

Mode of Evaluation:

- Internship Report: Students must submit a detailed report documenting their experiences and learnings.
- Presentation: A formal presentation reviewing the internship experience.
- Project Review: Evaluation based on the contributions and insights gained during the onsite project work.

Course Outcomes:

Upon successful completion of this internship, students will be able to:

CO1: Understand industry workflows, standards, and team dynamics through direct exposure to professional environments. (*L2: Understand*)

CO2: Apply technical knowledge and skills to real-world challenges encountered in industry projects. (*L3: Apply*)

CO3: Analyse the technical and non-technical aspects of industry projects, including economic, environmental, and societal impacts of engineering solutions. (*L4: Analyse*)

CO4: Evaluate personal and professional growth by reviewing learnings from industry engagement and assessing how these skills align with career goals. (*L5: Evaluate*)

CO5: Develop a detailed report and deliver a professional presentation summarizing internship experiences, contributions, and insights. (*L3: Develop*)